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Analytical Problem solving

1) Grey Level Transformation

a) Grey level transformation is used to improve image brightness and contrast, making important details more visible.

b) Negative transformation

$$s = 255 - r$$

Original (r) Transformed (s)

50 205

100 155

150 105

200 55

Interpretation: Bright areas become dark and dark areas become bright, producing a negative image.

2) Histogram Equalization

a) Histogram equalization improves image contrast by redistributing pixel intensities uniformly.

Intensity	Frequency	CDF
0	4	4
1	6	10
2	10	20
3	6	26

Total Pixels = 26

c) Equalized values (using $s = (L-1) \times \text{CDF}(N)$, where

$L=4$

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Intensity	Equalized value
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0	0
1	1
2	2
3	3

Interpretation: The pixel intensities are spread more evenly, resulting in better image contrast.

3) Gaussian Smoothing

a) Gaussian filtering reduces noise while preserving edges better than simple averaging because it gives more weight to the center pixel.

b) Kernel:

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Image:

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

Calculation: $\frac{1}{16} (10+40+10+40+160+40+10+40)$

New center pixel = $22.5 \approx 23$

Interpretation: Noise is reduced and the image

Image Smoother:

Edge Detection using sobel operator

Edge detection identifies boundaries of objects by detecting sudden changes in pixel intensity.

b) Sobel mask:

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Image:

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Gradient:

$$(-10 - 20 - 10) + (30 + 60 + 30) = 80$$

Interpretation: A strong horizontal edge is present.

5) Contrast stretching

a) Contrast stretching increases the intensity range of an image, making details clearer.

b) Formula:

$$S = \frac{(r - 50) \times 255}{150 - 50}$$

r s

50 0

100 $127.5 \approx 128$

150 255

Interpretation: The intensity range is expanded from 50-150 to 0-255, improving the image contrast.